PROJECT REPORT

NETWORK REQUEST MANAGEMENT

ServiceNow Service Catalog & Network Request Automation

PROJECT INFORMATION	DETAILS
Project Title	Automated Network Request & Fulfillment Management System
Student Name	Bantupalli Niharika
Technology	ServiceNow
Application	Network Request Management
Date	6 September 2026

SUBMITTED AS A PROJECT DOCUMENTATION REPORT

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Automated Network Request & Fulfillment Management System

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Project at a Glance

The project provides a centralized ServiceNow catalog-based process for employees to submit network service requests. It uses catalog variables, UI Policies, approval processing and Flow Designer automation to support request fulfillment.

CORE COMPONENT	ROLE
Service Portal / Service Catalog	Employee-facing request entry point
Catalog Variables & UI Policies	Capture information and guide form behaviour
Approval Process	Routes submitted requests for approval
Flow Designer	Automates workflow actions
Network Task	Tracks work assigned to the fulfillment team

01 INTRODUCTION

1.1 Project Overview

The Automated Network Request & Fulfillment Management System is a ServiceNow-based solution intended to provide employees with a centralized method for requesting network-related services. Employees access a Network Request Catalog Item through the Service Portal and enter the information required to process the request.

Catalog variables capture request details. Catalog UI Policies support dynamic form behaviour and mandatory-field validation. After submission, ServiceNow creates request records (REQ and RITM). The request is routed for approval; after approval, a Network Task is created and assigned to the Network Fulfillment Team for processing.

1.2 Purpose and Objectives

The purpose is to standardize network-service request submission and provide a traceable process from request entry to fulfillment. The project objectives are:

- Provide a single catalog entry point for network requests.
- Capture relevant and complete request information.
- Use form validation to reduce incomplete submissions.
- Route requests through an approval process.
- Create fulfillment tasks after approval.
- Assign tasks to the Network Team and improve request visibility.

1.3 Scope

The project scope includes the Service Catalog item, catalog variables, Catalog UI Policies, approval processing, Flow Designer automation, Network Task creation, group assignment and basic workflow validation. Large-scale concurrent-user load testing and advanced external integrations are outside the documented completed scope.

1.4 High-Level Workflow

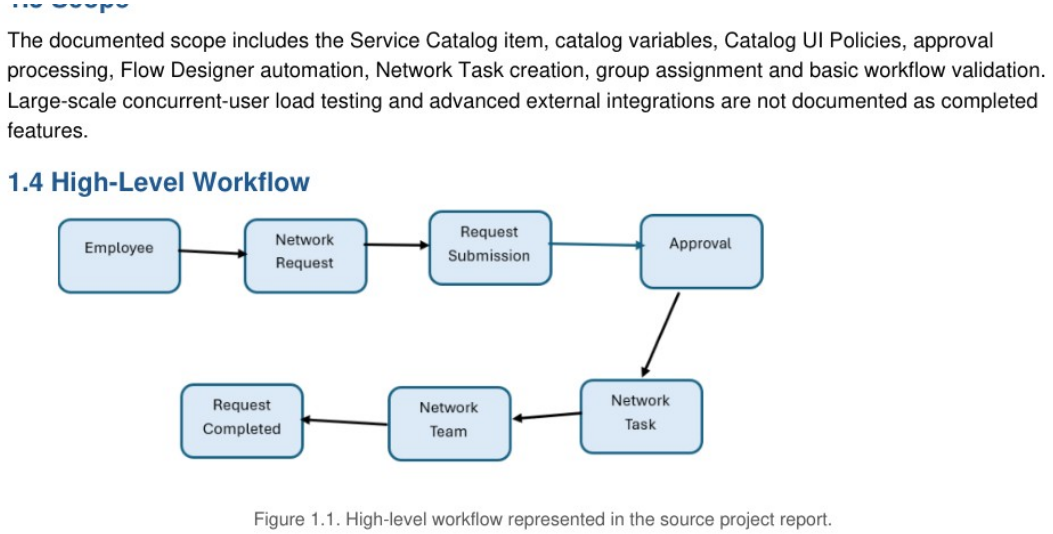


Figure 1.1. High-level workflow from employee request submission through approval and network fulfillment.

02 IDEATION PHASE

2.1 Problem Statement

Employees may experience difficulty when requesting network-related services if request channels and forms are unclear or unstructured. Missing information can lead to clarification cycles, while manual routing can delay approval and fulfillment. The project addresses this by providing a guided catalog item and a defined processing workflow.

PARAMETER	DESCRIPTION
User	Employees requesting network services
Problem	Network requests may be unclear or incomplete
Need	A simple and structured request process
Impact	Delays in approval and fulfillment
How might we?	How might we provide a centralized system for submitting and managing network requests efficiently?

2.2 Empathy Map – Employee Perspective

DIMENSION	EMPLOYEE PERSPECTIVE
Says	I need a simple way to request network services.
Thinks	I want a clear form with relevant options.
Does	Opens the portal, selects Network Request and submits details.
Feels	Wants fast and reliable network support.
Pain points	Incomplete requests, unclear forms and delayed processing.
Needs / goals	Simple submission, clear tracking and timely fulfillment.

2.3 Brainstorming and Prioritization

IDEA / FEATURE	PRIORITY
Network Request Catalog Item	High
Catalog Variables	High
Catalog UI Policies	High
Approval Process	High
Flow Designer Automation	High
Network Task Creation	High
Network Team Assignment	High
Request Status Tracking	Medium
Notifications	Medium
Reports and Dashboards	Medium

The high-priority features were selected for the initial implementation because they directly support the core request and fulfillment workflow.

03 REQUIREMENT ANALYSIS

3.1 Customer Journey Map

STAGE	EMPLOYEE ACTION	SYSTEM RESPONSE
Access	Opens Service Portal	Portal displays available services
Discover	Finds Network Request	Catalog item is displayed
Submit	Enters request details	Variables capture information
Validate	Completes required fields	UI Policies validate the form
Process	Submits request	REQ and RITM are created
Approval	Waits for approval	Approval workflow is triggered
Fulfillment	Request is processed	Network Task is created
Completion	Receives completed service	Request status is updated

3.2 Functional Requirements

ID	REQUIREMENT
FR-01	Employees shall be able to access the Network Request Catalog Item.
FR-02	The system shall capture the required request information.
FR-03	The system shall validate mandatory fields.
FR-04	Relevant fields shall be displayed dynamically where configured.
FR-05	The system shall create request records after submission.
FR-06	The request shall be sent for approval.
FR-07	A Network Task shall be created after approval.
FR-08	The Network Team shall process the assigned task.

3.3 Non-Functional Requirements

ID	REQUIREMENT
NFR-01	The system should provide a simple user interface.
NFR-02	The system should use ServiceNow access controls.
NFR-03	The workflow should process requests consistently.
NFR-04	The system should provide normal ServiceNow response performance.

04 DATA FLOW AND TECHNOLOGY STACK

4.1 Logical Data Flow Diagram

4. DATA FLOW AND TECHNOLOGY STACK

4.1 Logical Data Flow Diagram

NFR-02	The system should use ServiceNow access controls.
NFR-03	The workflow should process requests consistently.
NFR-04	The system should provide normal ServiceNow response performance.

3.3 Logical Data Flow Diagram



Figure 4.1. Logical data flow diagram reproduced from the supplied project report. It depicts the employee, portal, catalog item, validation, approval, request records, task and fulfillment team interactions.

4.2 Technology Stack

No.	Component	Purpose in this project	Technology / record
1	User interface	Employee-facing request interface	Service Portal / Service Catalog
2	Catalog	Network request submission	Service Catalog
3	Rules	Dynamic fields and validation	Catalog UI Policies
4	Automation	Approval and workflow processing	Flow Designer
5	Data	Request and fulfillment records	REQ / RITM / Tasks

Figure 4.1. Logical data flow showing employee, portal, catalog item, validation, approval, request records, task and fulfillment team interactions.

4.2 Technology Stack

NO.	COMPONENT	PURPOSE IN THIS PROJECT	TECHNOLOGY / RECORD
1	User interface	Employee-facing request interface	Service Portal / Service Catalog
2	Catalog	Network request submission	Service Catalog
3	Rules	Dynamic fields and validation	Catalog UI Policies
4	Automation	Approval and workflow processing	Flow Designer
5	Data	Request and fulfillment records	REQ / RITM / Tasks
6	References	User and group information	sys_user / sys_user_group

4.3 Data and Record Notes

The system uses REQ and RITM request records and Network Tasks as the principal records involved in the documented workflow. It also uses sys_user and sys_user_group as reference data for users and assignment groups.

05 PROJECT DESIGN AND ARCHITECTURE

5.1 Problem–Solution Fit

PARAMETER	PROJECT FIT
Problem	Employees need a simple and structured way to submit network requests.
Solution	A ServiceNow-based Network Request Management system.
Customer benefit	Guided request entry and better request visibility.
Operational benefit	Structured requests improve approvals and fulfillment.
Uniqueness	Combines Service Catalog, UI Policies, approvals and task automation.
Scalability	Additional request types and automation can be added later.

5.2 Proposed Solution

Employees submit network-related requests through a Network Request Catalog Item. Catalog variables capture request information, while Catalog UI Policies provide dynamic form behaviour and validation. After submission, the request is processed through an approval workflow. Once approved, a Network Task is created and assigned to the Network Fulfillment Team.

5.3 Solution Architecture

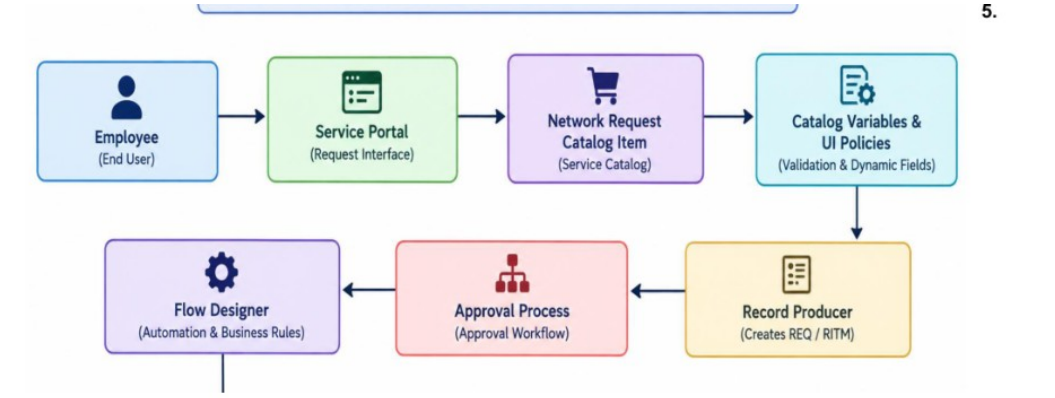


Figure 5.1. Solution architecture showing the Service Portal, catalog item, variables and UI policies, record creation, approval, Flow Designer, Network Task and fulfillment team.

5.4 Workflow Explanation

- Employee opens the Service Portal and selects the Network Request catalog item.
- The employee completes the catalog variables; configured UI Policies guide input and validation.
- The submitted request creates the associated REQ and RITM records.
- The approval process evaluates the request.

Figure 5.1. Solution architecture connecting the Service Portal, catalog item, variables and UI policies, record creation, approval, Flow Designer, Network Task and fulfillment team.

5.4 Workflow Explanation

- Employee opens the Service Portal and selects the Network Request catalog item.
- The employee completes the catalog variables; configured UI Policies guide input and validation.
- The submitted request creates the associated REQ and RITM records.
- The approval process evaluates the request.
- Following approval, the automation creates a Network Task and assigns it to the Network Team.
- The fulfillment team processes the task and the request is updated to its completed state.

06 PROJECT PLANNING AND SCHEDULING

6.1 Sprint Plan

SPRINT	FUNCTIONAL AREA	USER STORY / TASK	POINTS	PRIORITY	ASSIGNED STUDENT
1	Foundation	Create Network Request components and fields	6	High	Bantupalli Niharika
2	Catalog & rules	Create catalog item, variables and UI Policies	6	High	Bantupalli Niharika
3	Automation	Configure approvals and Network Task creation	6	High	Bantupalli Niharika
4	Testing	Testing and documentation	2	High	Bantupalli Niharika

6.2 Implementation Activities

The project plan is organized into four sequential work areas. The foundation sprint establishes the request components and fields. The catalog and rules sprint configures the user-facing item, variables and UI Policies. The automation sprint configures approval and task creation. The final sprint focuses on testing and documentation.

6.3 Planning Notes

The sprint plan identifies the functional areas, tasks, story points, priorities and assigned student. Sprint dates and actual completion dates are not specified.

07 FUNCTIONAL TESTING AND USER ACCEPTANCE TESTING

7.1 Functional Testing

TEST ID	SCENARIO	EXPECTED RESULT	STATUS
TC-01	Open Service Portal	Portal loads successfully	Pass
TC-02	Find Network Request	Catalog item is visible	Pass
TC-03	Open request form	Form opens successfully	Pass
TC-04	Enter request details	Variables are available	Pass
TC-05	Select device type	Relevant options display	Pass
TC-06	Validate fields	Mandatory validation works	Pass
TC-07	Submit request	REQ and RITM are created	Pass
TC-08	Approval process	Request goes for approval	Pass
TC-09	Create Network Task	Task is created	Pass
TC-10	End-to-end test	Complete workflow works	Pass

7.2 User Acceptance Testing (UAT)

UAT ID	SCENARIO	ACCEPTANCE CRITERIA	RESULT
UAT-01	Open Service Portal	Portal accessible	Accepted
UAT-02	Find Network Request	Catalog item visible	Accepted
UAT-03	Open form	Request form opens	Accepted
UAT-04	Select Device Type	Option selectable	Accepted
UAT-05	Validate fields	Required validation works	Accepted
UAT-06	Submit request	Request created	Accepted
UAT-07	Approval	Approval works	Accepted
UAT-08	Task creation	Task assigned	Accepted
UAT-09	Complete request	Request completed	Accepted

7.3 UAT Conclusion

The functional test cases are recorded as Pass and the UAT cases are recorded as Accepted.

08 PERFORMANCE TESTING AND RESULTS

8.1 Performance Testing

The implemented workflow was tested for normal interactive usage within the ServiceNow Personal Developer Instance. The application successfully handled the following activities:

- Catalog form loading.
- Request submission.
- Validation processing.
- Approval workflow execution.
- Network Task creation.

Testing limitation: Large-scale concurrent-user load testing was outside the scope of the project.

8.2 Performance Evidence and Reporting

Measured response times, throughput, concurrent-user counts, CPU or memory utilization, and load-test tool output are not included. Therefore, no numerical performance claim is made in this report.

8.3 Results Summary

AREA	RESULT
Catalog item	Created
Catalog variables	Required fields configured
UI Policies	Dynamic behaviour configured
Approval process	Approval configured
Flow Designer	Workflow automation configured
Network Tasks	Tasks created for fulfillment
Service Portal	Catalog accessible
Testing	Workflow validated

The core workflow components were configured and validated in the project environment.

09 OUTPUT SCREENSHOTS

ServiceNow configuration and employee request interface

9.1 Network Request Catalog Item

9. OUTPUT SCREENSHOTS

9.1 Network Request Catalog Item

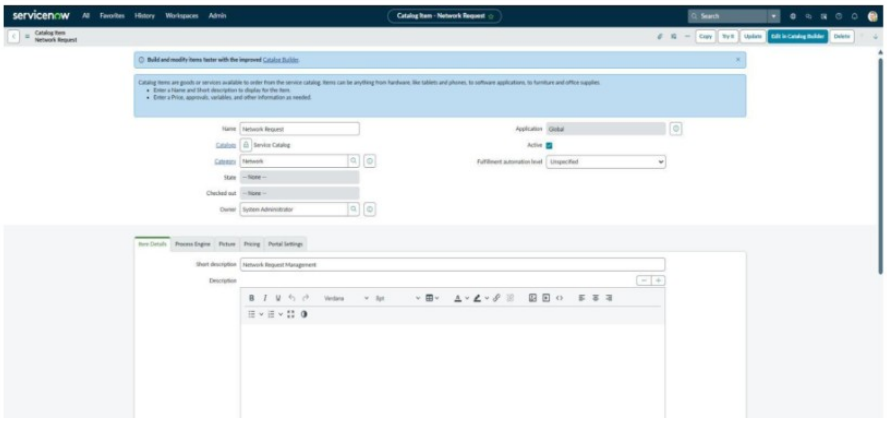


Figure 9.1. Network Request Catalog Item configuration screen.

9.2 Network Request Form



Figure 9.1. Network Request Catalog Item configuration screen from the supplied report.

9.2 Network Request Form

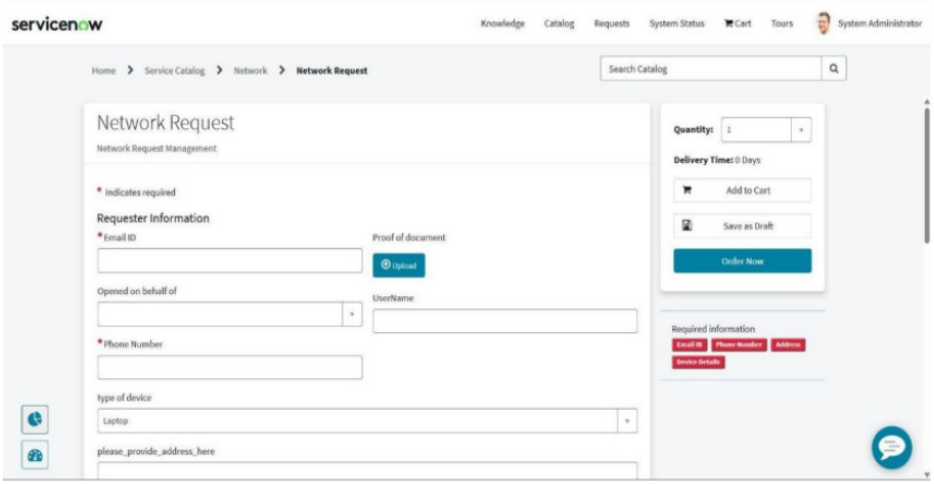


Figure 9.2. Employee-facing Network Request form showing request information fields and the order action.

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09 OUTPUT SCREENSHOTS – CONTINUED

Request submission and notification evidence

9.3 Submitted Network Request Record

9. OUTPUT SCREENSHOTS (CONTINUED)

9.3 Submitted Network Request Record

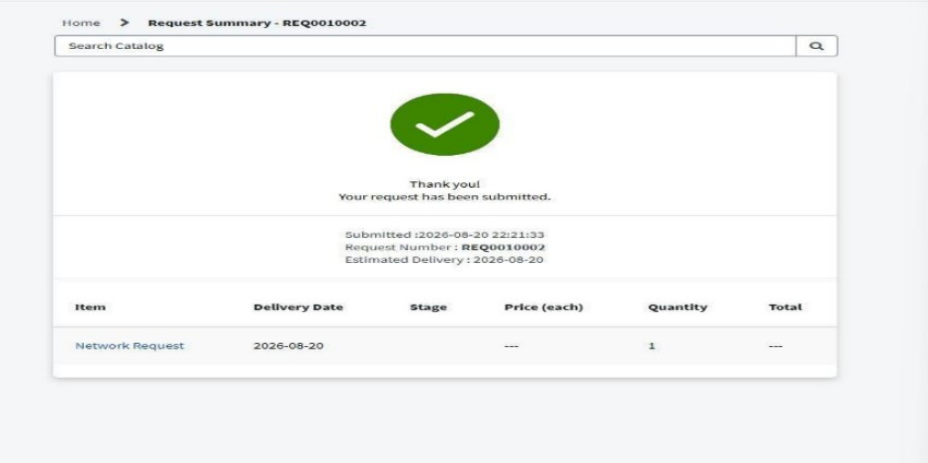


Figure 9.3. Request submission confirmation and summary record.

9.4 Email Notification

Figure 9.3. Request submission confirmation and summary record shown in the supplied screenshot.

ail Notification



Figure 9.4. Email notification preview included in the original project report.

ıts are retained from the original PDF. Before final HR submission, ensure that the screenshots are readable and do not
rsonal, organizational or instance credentials.

Figure 9.4. Email notification preview.

10 WORK ITEM VALIDATION, ADVANTAGES AND LIMITATIONS

10.1 Work Item Status Validation

WORK ITEM	STATUS	VALIDATION
Network Request Catalog	Completed	Catalog item created
Catalog Variables	Completed	Required fields configured
UI Policies	Completed	Dynamic behaviour configured
Approval Process	Completed	Approval configured
Flow Designer	Completed	Workflow automation configured
Network Tasks	Completed	Tasks created for fulfillment
Service Portal	Completed	Catalog accessible
Testing	Completed	Workflow validated

10.2 Advantages

- Provides centralized network request submission.
- Reduces incomplete request information.
- Improves request standardization.
- Automates approvals and task creation.
- Improves visibility into the request lifecycle.
- Reduces manual effort for the Network Team.

10.3 Limitations / Disadvantages

- Depends on correct ServiceNow configuration.
- Initial configuration requires platform knowledge.
- Complex workflows may require additional maintenance.
- Advanced integrations may require additional development.
- Large-scale concurrent-user performance was not tested within the documented scope.

11 CONCLUSION AND FUTURE SCOPE

11.1 Conclusion

The Automated Network Request & Fulfillment Management System demonstrates a structured approach to managing network service requests using ServiceNow. Employees can submit requests through a centralized catalog, provide required information, and benefit from configured validation and approval processing. Following approval, the workflow creates a Network Task for fulfillment by the Network Team.

The project brings together ServiceNow Service Catalog, Catalog UI Policies, Flow Designer, approval processing and task management to support a more consistent request lifecycle. The test outcomes and screenshots provide evidence for the implemented scope.

11.2 Future Scope

- Email and mobile notifications.
- Advanced dashboards and reporting.
- Service Level Agreement (SLA) monitoring.
- Additional network service categories.
- Automatic priority calculation.
- Integration with external network monitoring tools.
- AI-based request classification.
- Virtual Agent support.
- Knowledge articles for common network issues.

12 APPENDIX AND REFERENCES

Appendix A – Technology Components

NO.	COMPONENT
1	ServiceNow Service Catalog
2	Catalog Items
3	Catalog Variables
4	Catalog UI Policies
5	Flow Designer
6	Approval Engine
7	REQ and RITM Records
8	Network Tasks
9	sys_user
10	sys_user_group

Appendix B – Project Flow

- Catalog UI Policies
- Flow Designer
- Approval Engine
- REQ and RITM Records
- Network Tasks
- sys_user
- sys_user_group

Appendix B – Project Flow

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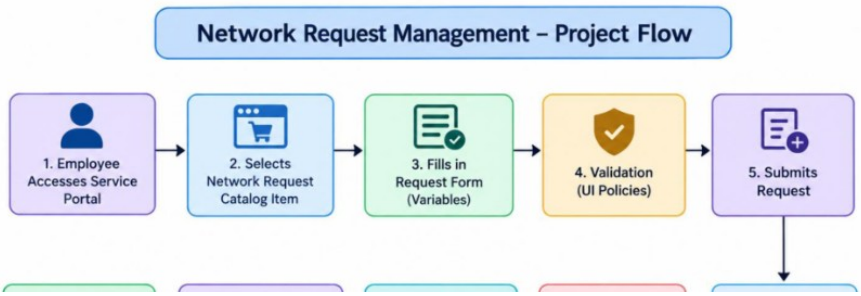


Figure 12.1. End-to-end project flow.

Appendix C – Acronyms

TERM	MEANING
REQ	Request record
RITM	Requested Item record
UAT	User Acceptance Testing
UI	User Interface
SLA	Service Level Agreement
DFD	Data Flow Diagram